

EDUCATION

Master's - Master of Applied Computer Science

Jan 2024 - Jan 2026

CONCORDIA UNIVERSITY, MONTREAL, QUEBEC, CANADA

- **Relevant Courses:** Distributed System Design | Natural Language Processing | Machine Learning | Artificial Intelligence | Programming and Problem Solving.

SKILLS

Programming: Java | Python | SQL | PySpark.

AI: Natural Language Processing (NLP) | Transformers | LSTMs | LLMs.

Data Science & Analytics: Pandas | NumPy | Scikit-Learn | TensorFlow | PyTorch.

Tools: AWS | Git | IntelliJ | VS Code.

SoftSkills: Communication | Time-management | Teamwork | Adaptability | Task Prioritization | Problem solving.

PERSONAL AND ACADEMIC PROJECTS

Machine Generated Text Detection (2024)

- The aim of the project is to distinguish whether the generated text is Machine generated or Human using unidirectional and bidirectional LSTMS.
- Preprocessed and trained on a **610,767-sample** dataset, **70.18% accuracy / 0.6818 Macro-F1** with **UniLSTM** (loss **0.4210**) and **70.11% accuracy / 0.6772 Macro-F1** with **BiLSTM** (loss **0.3077**), showing clear comparison.
- **SKILLS:** Python | Pytorch | Nltk | Pandas | Sklearn | Machine Learning | [GitHub](#)

Machine Translation with Transformers (2024)

- Developed a machine translation pipeline using Hugging Face Transformers on the **IWSLT 2017 English French dataset** on a **100,000-sample** dataset over **10 epochs**.
- Implemented an end-to-end pipeline covering **data preprocessing, training, hyperparameter tuning, and evaluation**.
- **Custom Transformer** achieved **0.8581 Precision / 0.8435 Recall / 0.8506 F1 / 0.2978 METEOR**; **T5** achieved **0.8960 Precision / 0.8987 Recall / 0.8972 F1 / 0.5160 METEOR**, highlighting performance gains from trained model.
- **SKILLS:** Python | Hugging Face Transformers | NLP | Machine Translation | Dataset Preprocessing | BERT Score | ROUGE | Model Evaluation | Text Normalization | [GitHub](#)

Multi-Instance Iris Template-Based Intruder Detection Using Minkowski Distance (2023)

- The aim of the project is distinguished whether the user is intruder or valid user based on human Iris.
- Achieved a high **accuracy rate of 98.75%** by utilizing the Minkowski distance technique compared to existing Hamming distance models.
- Integrated Daugman's algorithm for iris localization and segmentation, template generation via Logical OR operation, key generation using XOR operations on primary diagonals, and classification via the K-Nearest Neighbor (KNN) algorithm.
- **SKILLS:** Python | Machine-learning | [Publication](#)

Stock Market Price Prediction (2022)

- The web application takes data from Yahoo Finance and shows the predicted closing and original prices.
- **SKILLS:** Streamlit | Python | TensorFlow | HTML | Keras | Sqlite | Machine Learning | [GitHub](#)

WORK HISTORY AND CERTIFICATIONS

Python Programming Trainee (APSSDC)

Mar 2022 - Jun 2022

- Completed a structured internship focused on Python programming, data structures, and real-time project development.
- By using OOP, data processing techniques, **Linux CLI** and **Bash** to automate deployments, and operational tasks.
- Strengthened problem-solving and debugging skills while gaining insight into professional software development workflows.

Aws Cloud Intern (AICTE)

Oct 2021 - Dec 2021

- Deployed and managed scalable cloud infrastructure using core AWS services including EC2, S3, and Lambda.
- Gained hands-on experience with IAM, VPCs, and serverless architecture for secure and efficient operations.
- Designed and implemented a sample cloud-based web application demonstrating cost-optimized deployment.

Certifications: AWS Solutions Architect | AWS Cloud Practitioner | Wipro Talent Next (WIPRO) | SQL (HACKER RANK).